

☀ IFS AWARDS 2026 · CHALLENGE SOLUTION PROPOSAL · BRITE SOLAR

SMDC-Agri

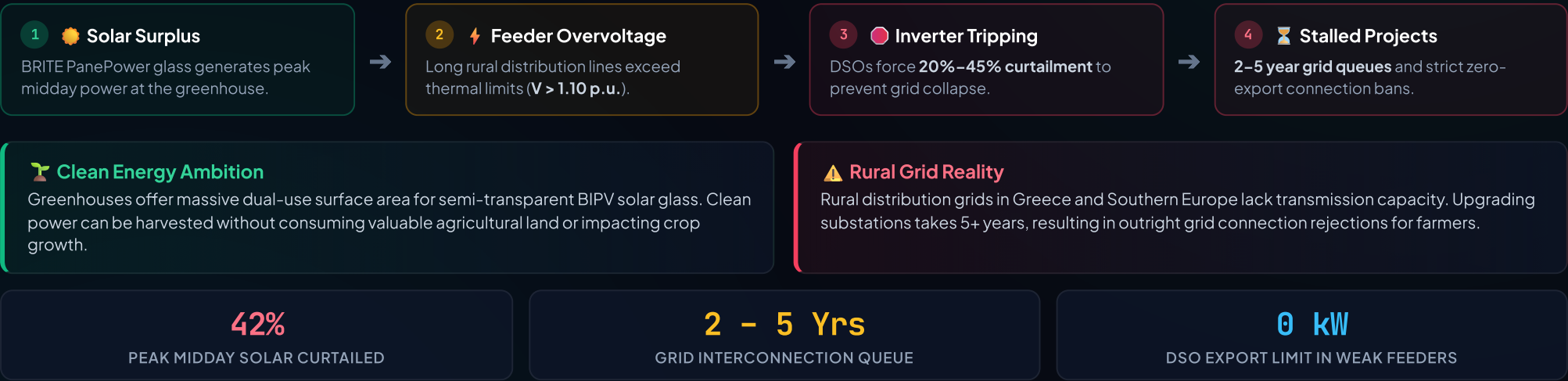
Integrating Agri-PV Systems in Weak Rural Grids via Behind-The-Meter
Sovereign Edge Compute & Dynamic Load Shedding



Agri-PV Greenhouse with Semi-Transparent Solar Glass & Sovereign Edge Datacenter

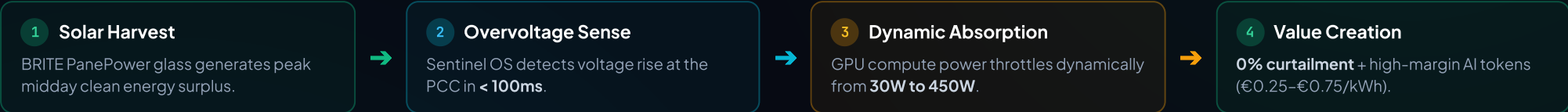
The Rural Grid Bottleneck for Agri-PV Expansion

Why high-potential agrivoltaic projects face grid connection refusal, severe curtailment, and stalled licensing



The Core Innovation: "Compute as a Virtual Battery"

Transforming stranded solar kilowatt-hours into on-premise AI intelligence in sub-100 milliseconds



💡 Why Compute Outperforms Chemical Batteries (BESS) Alone:

Chemical batteries suffer from high upfront CapEx, finite charge cycle degradation, and fire risk. **SMDC-Agri** absorbs stranded power behind the meter into computing work—yielding **10x higher economic productivity per kWh** while guaranteeing 100% compliance with DSO zero-export limits.

Zero Degradation · Immediate Digital Revenue

< 100ms

LOAD STEP RESPONSE

< 2%

SOLAR CURTAILMENT RATE

> 95%

SELF-CONSUMPTION RATE

4.2 Yrs

PAYBACK PERIOD (VS. 8.5 YRS)

Architecture & Zero-CapEx Validation Engine

Validating sub-second dynamic load shedding on in-house DGX Spark hardware without equipment purchasing



Tier 1: Power & Thermal Plane

Hardware Layer

Direct DC coupling with BRITE PanePower glass + 48V LiFePO4 buffer + closed-loop liquid cooling rejected into greenhouse thermal mass.

Tier 2: Sentinel OS Orchestrator

Real-Time Shedding

Hardware priority enforcement: guarantees **L₀ Life Support** (irrigation, ventilation) while dynamically throttling **L₂-L₄ AI compute**.

Tier 3: In-House DGX Spark Platform

Zero-CapEx Validation

Hardware-in-the-Loop (HIL) emulation with real PanePower I-V curves, simulating feeder voltage sags and sub-100ms load shedding.

€0 CapEx

HARDWARE VALIDATION SPEND

30W → 450W

DYNAMIC LOAD RANGE

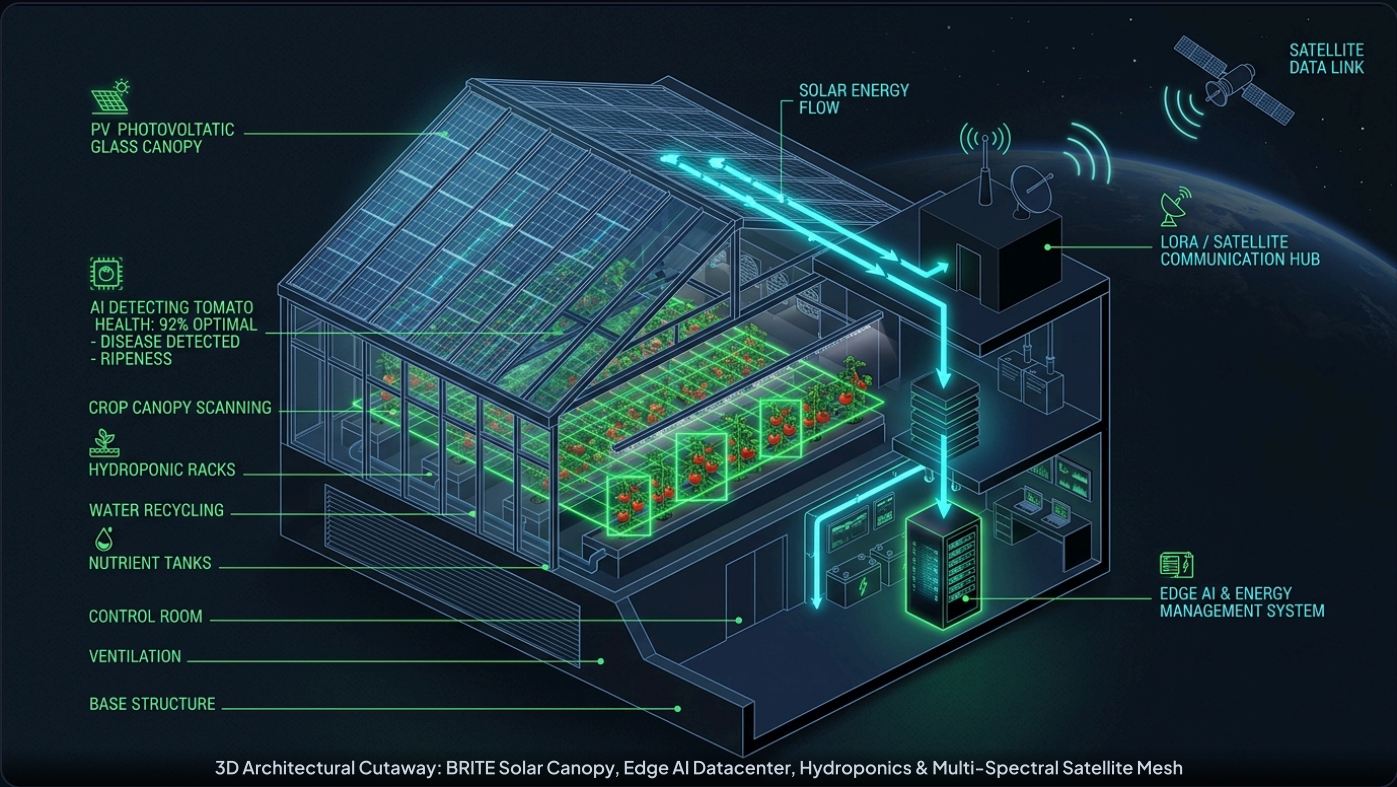
100%

AIR-GAPPED SOVEREIGN SECURITY

Greenhouse Digital Twin & Smart Edge AI

4. EDGE APPLICATIONS

Complete 3D architecture: real-time canopy computer vision, PAR optimization, and autonomous hydroponics



1. Edge Computer Vision

TensorRT models detect fungal mold, leaf blight, and fruit ripeness locally. Zero cloud egress costs and instant alerts.

2. Dynamic PAR Balancing

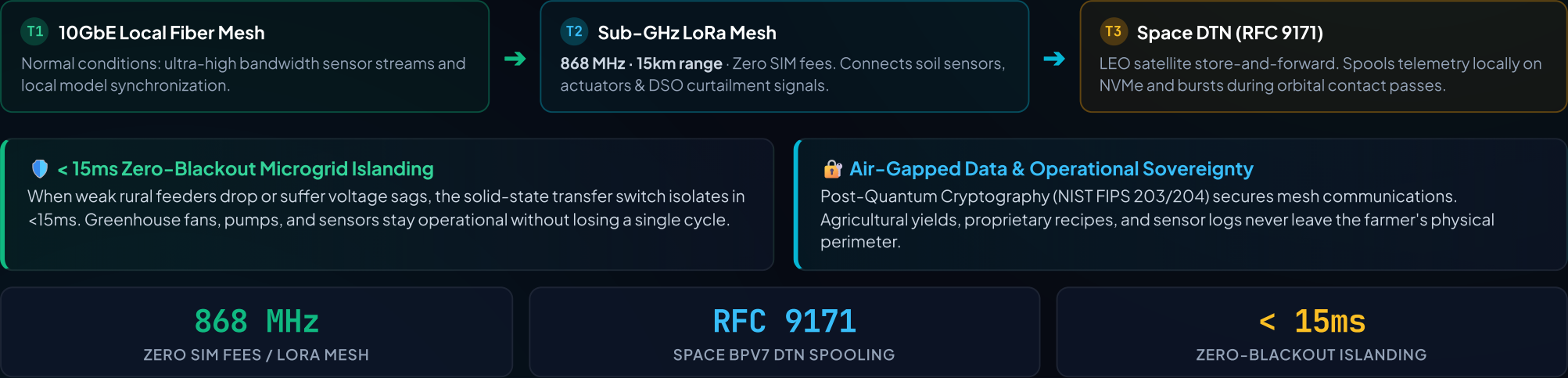
Couples live solar radiation with plant photobiology, balancing energy harvest vs. crop Daily Light Integral (DLI).

3. Autonomous Life-Support

Guarantees un-sheddable power for nutrient dosing, water pumps, and ventilation fans during all weather conditions.

Off-Grid Resilience & Multi-Spectral Comms

Autonomous operation in remote rural regions with zero reliance on public cellular or commercial fiber networks



Quantified Techno-Economics: 3-Year ROI Payback

Turning regulatory gridlocks into a high-margin dual-revenue commercial deployment model

METRIC / PARAMETER	STANDARD AGRI-PV (WEAK RURAL GRID)	AGRI-PV + SMDC-AGRI (BEHIND-THE-METER)	NET ADVANTAGE / IMPACT
Grid Connection Status	Denied or delayed by DSO (overvoltage risk)	Approved (Zero-Export / Dynamic Limit compliant)	Fast-tracks project licensing
Peak Solar Curtailment	25% – 45% peak midday energy wasted	< 2% (Absorbed by local AI compute)	+35% energy recovery
Revenue per Curtailed kWh	€0.00 / kWh (dumped or curtailed)	€0.25 – €0.75 / kWh (AI compute productivity)	High-margin digital revenue
Upfront Validation CapEx	Uncertain / High Risk	€0 (Utilizes existing DGX Spark testbed)	Zero financial barrier for POC
Capital Payback Period	8.5 – 9.5 Years (curtailment penalty)	3.8 – 4.2 Years (subsidized by compute)	50%+ Faster Capital Payback

+€18,400 / yr

ADDED COMPUTE REVENUE (250KWP)

€0 CapEx

VALIDATION HARDWARE PURCHASE

100%

BRITE GLASS COMPATIBILITY

The Journey to Impact: Bootcamp & Validation Roadmap

Focused technical co-design and validation milestones with BRITE using our in-house DGX Spark testbed

Bootcamp Stage

Direct technical co-design with BRITE engineers on PanePower I-V curves, glass specs & DSO constraints.

Validation Stage

HIL Validation on DGX Spark: Sub-100ms dynamic load shedding demonstrated on live testbed.

The Multidisciplinary Engineering Team:

Metatopia Studio · Thessaloniki, Greece

Ilias Chrysovergis (CEO/CTO, Dipl. ECE AUTH, MSc Imperial College London, PhD cand., Global 1st Microsoft Imagine Cup) · **Archontis Kostis** (Lead Software & Cloud Architect, MSc UoM) · **Elena Vagiannopoulou** (Operations & Materials Lead, BSc Chemistry AUTH) · **Nikos Nikolaidis** (Full Stack & Systems, BSc UoM) · **Stelios Palatidis** (Edge & Mobile Systems, BSc UoM).

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